AI-GNITION 2026 · PROBLEM STATEMENT 1

Nest Helper

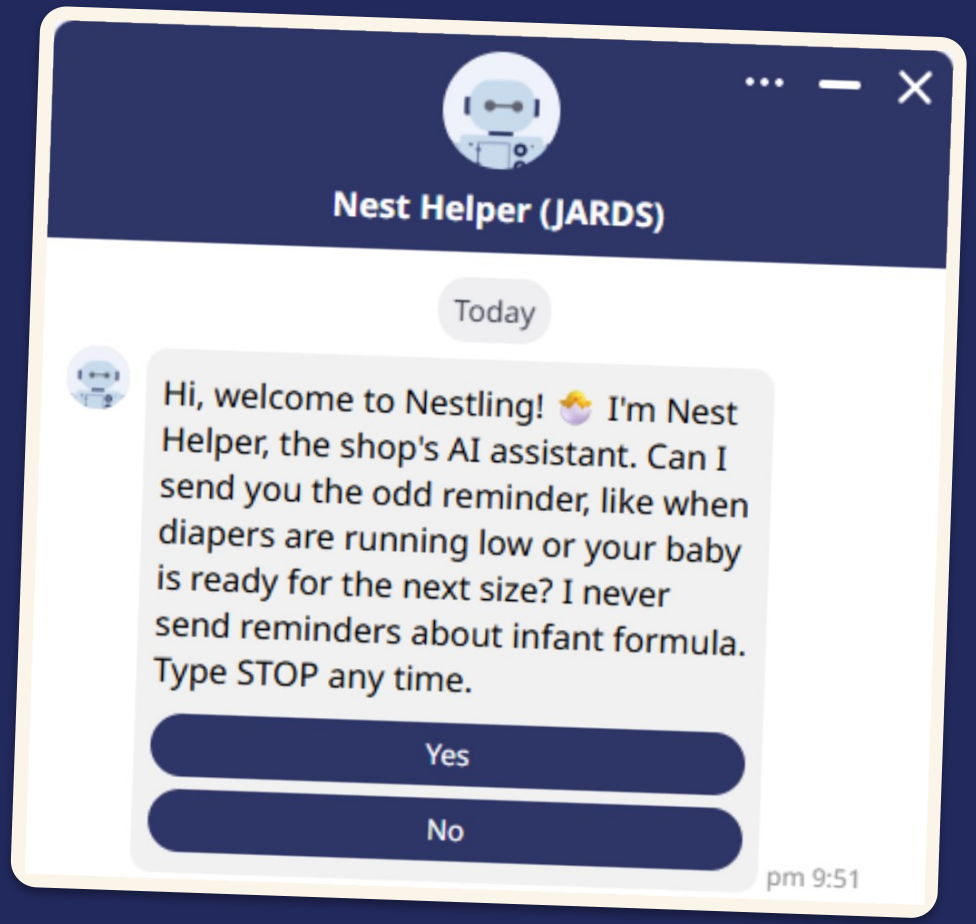
Smart size, perfect time, zero stress.

A chatbot for Nestling, a Singapore baby shop. It reads each family's orders and picks one useful reminder, or nothing at all.

Team JARDS

Jeiyanth · Devanshi · Adhvik · Shivangi · Reyansh

NPS International School Singapore



the first thing every parent sees

Nestling is a fictional shop. All customers and orders are simulated.

THE PROBLEM

2 a.m.

The last diaper is on the baby. The shop's latest message is the same sale it sent to everyone.

Babies change on a schedule. In the first 18 months, from our shop's product data:



5 diaper sizes, 6 clothing sizes, a new pack every 7 to 12 days. Miss the moment and a parent may buy elsewhere.

Who has it: parents of babies from newborn to 3 years in Singapore, and the small baby shops that want them back.

OUR ANSWER

One message for one baby, only when it’s due.

Many shops

One sale, sent to every customer, whenever the shop wants to sell.

Nest Helper

One message for that baby’s age and orders, only if the parent said yes, with a reason it can show.

1 Read → **2 Decide** → **3 Check** → **4 Write** → **5 Act**

- **1 Read** — orders, baby age, consent
- **2 Decide** — refill, size-up, milestone or win-back
- **3 Check** — consent, 7-day gap, formula rules
- **4 Write** — words for that baby
- **5 Act** — reorder, wait, stop or ask a person

Sometimes the best message is no message. 7 of our 20 families get none.

THE DECISION ENGINE

Safety checks first, then one journey.

Each family, one at a time

→ **Check 1**
Agreed to reminders?
No: stay quiet. C05, C20

→ **Check 2**
Only infant formula due?
Yes: stay quiet. C07

→ **Check 3**
Snoozed, or messaged in the last 7 days?
Yes: wait. C09

All clear? The first journey that fits wins, in this order:

1 Refill
From the family's own reorder gap, diapers, wipes or milk run out between 2 days ago and 3 days from now.

2 Size-up
The baby is within 1 month of the top of the last clothing size bought.

3 Milestone
The baby is 6 or 12 months old.

4 Win-back
2 or more orders, and 60 or more days since the last one.

Nothing fits: no message (C12, C13, C17). Otherwise the engine picks exactly one message, with its own "Why this message?" pack. Code: engine/decision_engine.py

THE RESULT

We ran it on 20 families. 13 get a message. 7 hear nothing.

C01 10m **Refill**	**C02** 2m **Size-up**	**C03** 12m **Milestone**	**C04** 14m **Win-back**	**C05** 9m **No consent**
C06 4m **Refill**	**C07** 4m **Formula only**	**C08** 15m **Refill (milk)**	**C09** 8m **Cooldown**	**C10** 1m **Size-up**
C11 11m **Refill**	**C12** 7m **Nothing due**	**C13** 23m **Nothing due**	**C14** 18m **Win-back**	**C15** 3m **Refill**
C16 6m **Milestone**	**C17** 5m **Nothing due**	**C18** 13m **Win-back**	**C19** 9m **Refill**	**C20** 16m **No consent**

13 messaged

Refill 6 · Size-up 2
Milestone 2 · Win-back 3

7 held back

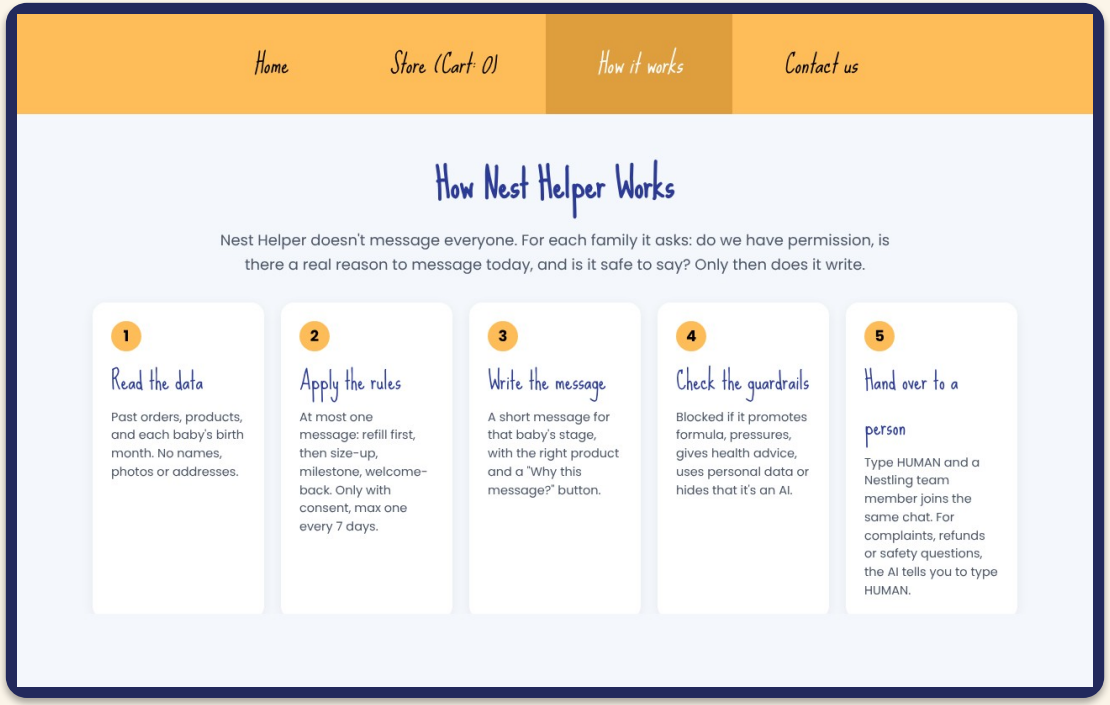
No consent 2 · Infant formula 1
Cooldown 1 · Nothing due 3

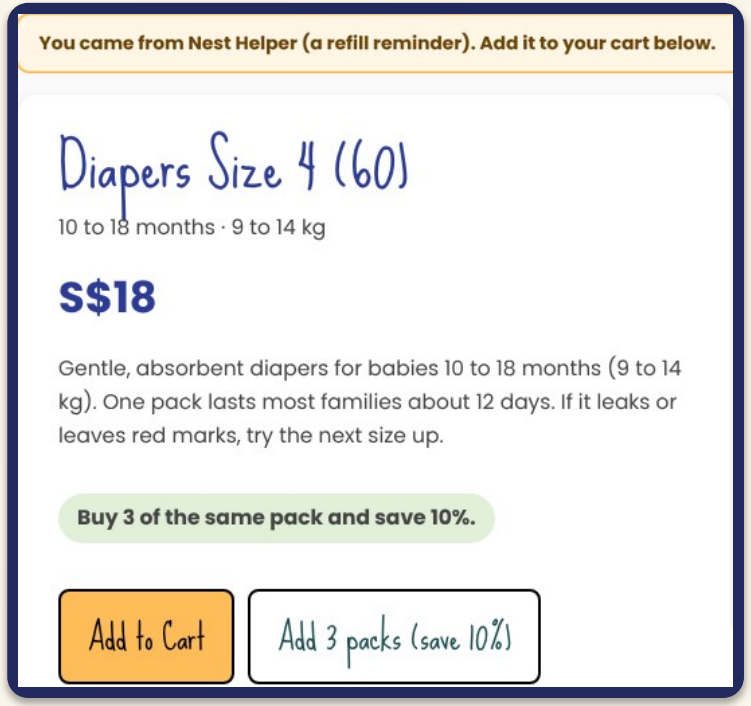
C03 fits two journeys. Milestone wins, win-back waits.

Fictional families. Real output of our engine for 20 Oct 2026 (output/decisions.csv). Age in months.

THE TEST SHOP

Nestling: a fictional shop, built to test it end to end.





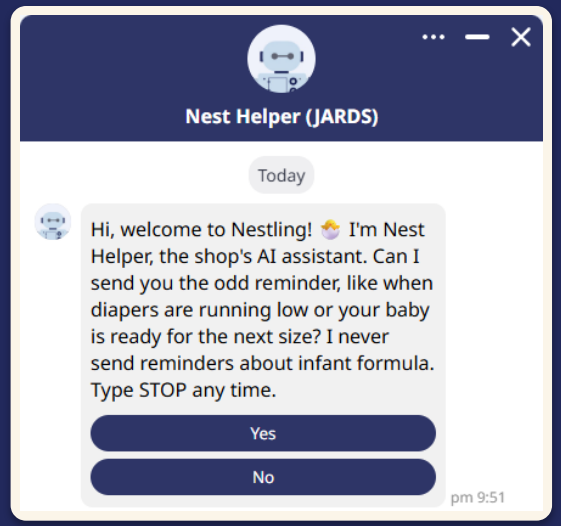
a tagged link shows the parent came from the bot

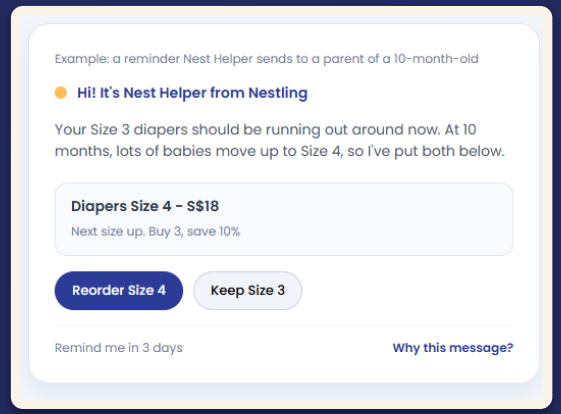
- 26 product pages: diapers in every size, clothes, Stage Bundles, toys, formula and milk
- Buy 3 packs of the same diapers or wipes and save 10%

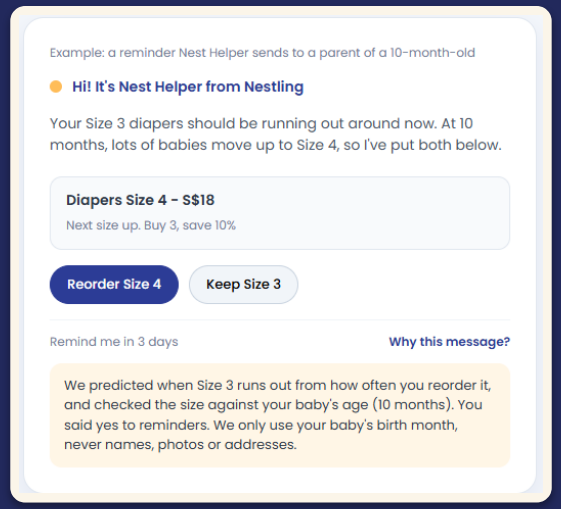
- Checkout is switched off, because the shop is fictional
- The How it works page explains the five steps

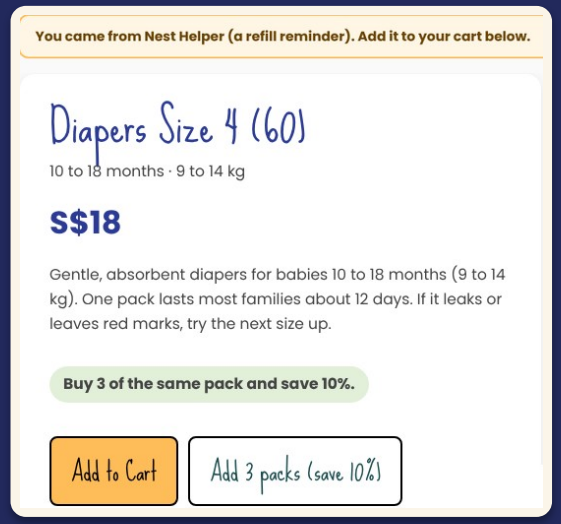
HOW THE PROTOTYPE WORKS

What a parent sees, on our live website and bot.









1 Says yes

The bot says it is an AI assistant and asks first.

2 Gets one reminder

Written for that baby, with one-tap buttons.

3 Asks why

"Why this message?" shows the data used.

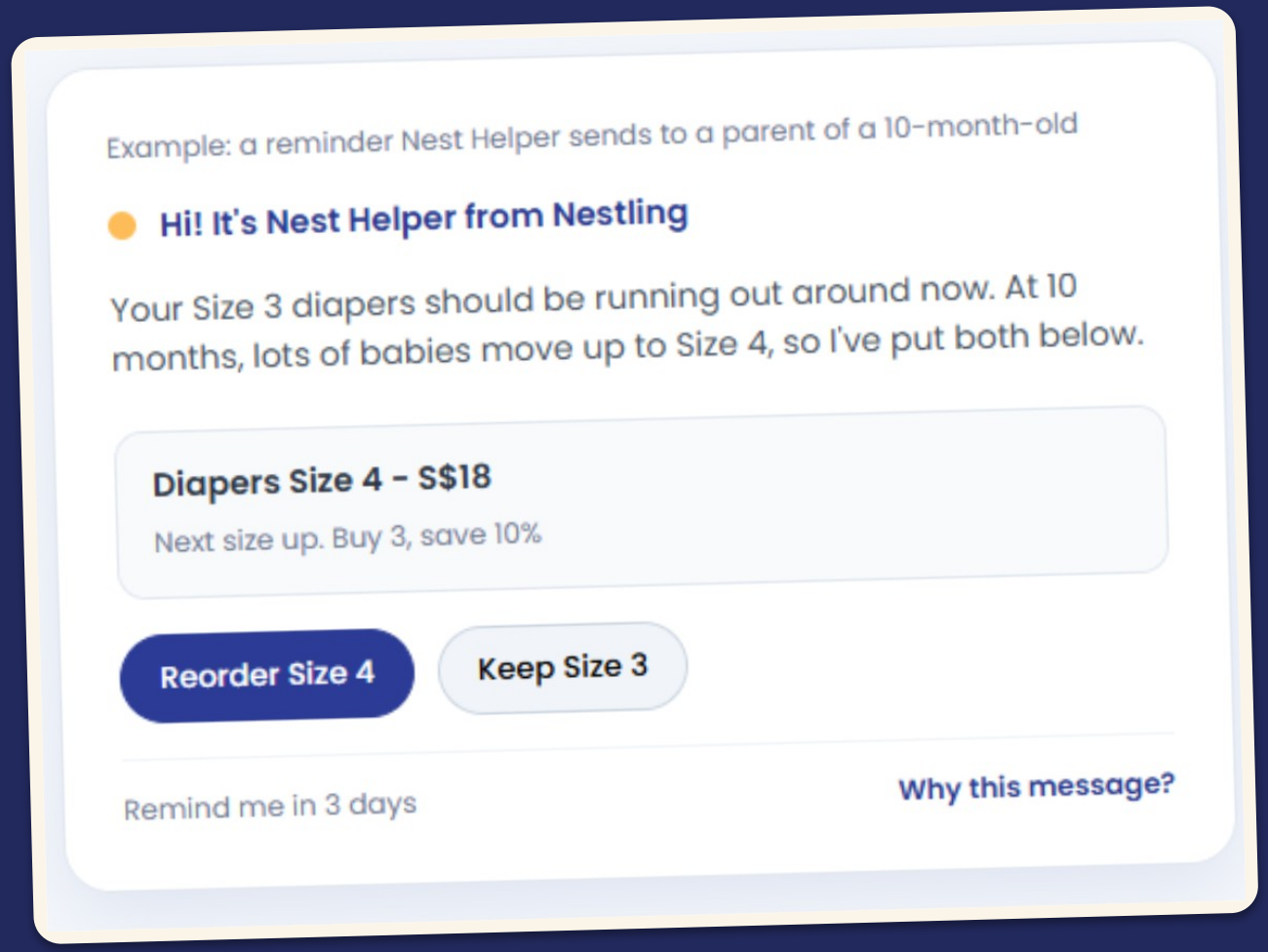
4 Lands on the product

The button opens that product page, tagged to the reminder.

Any time: STOP ends reminders · HUMAN brings in a person · ASK gets AI answers from our shop facts

ONE FAMILY, CLOSE UP

C01: a 10-month-old on Size 3 diapers.



Size 3 is due to run out
worked out from how often this family reorders

Many 10-month-olds move up
so Size 4 is offered first

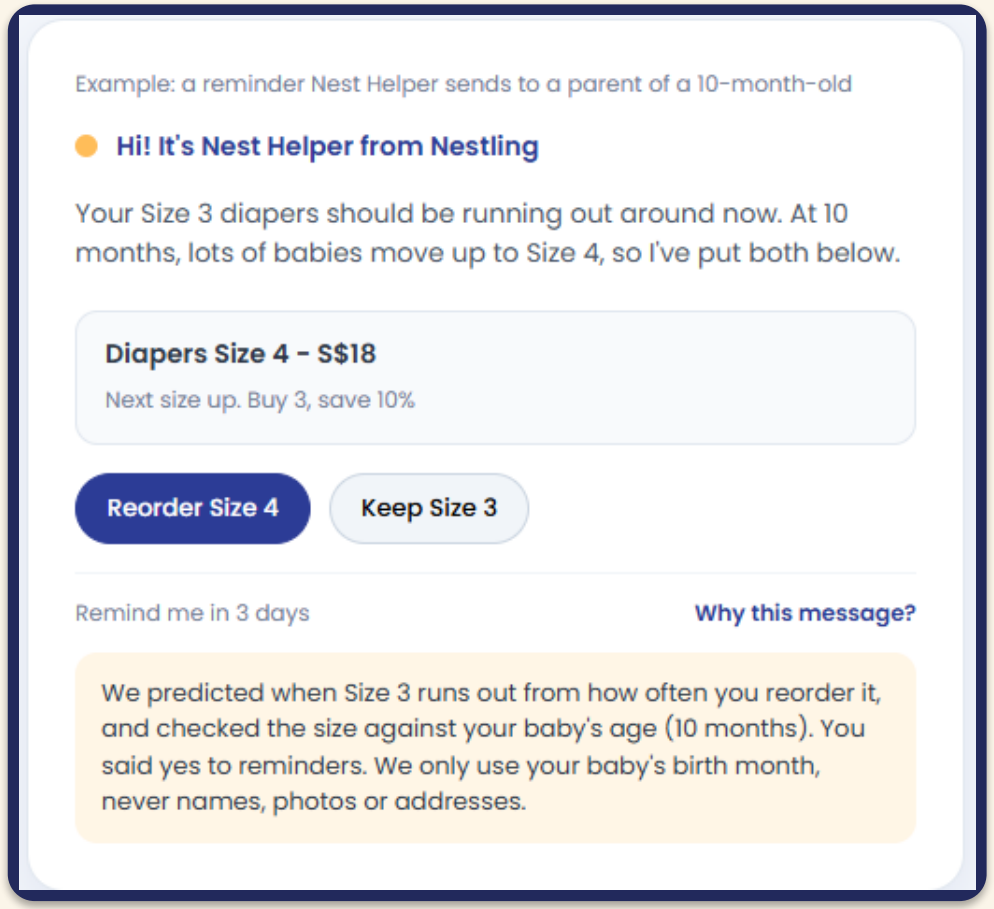
Not ready? Keep Size 3
the parent decides, not the bot

Buy 3, save 10%
a stock-up deal, and no deals ever on formula

same wording as live bot pack RF-DP3-UP-DP4

TRANSPARENCY

Tap “Why this message?” and the bot shows its data.



Used	**When you last ordered and how often you reorder**
Used	**Your baby’s age band, to check the size**
Used	**That you said yes to reminders**
Never used	**Your name, photos or address**

every reminder has one of these

THE OTHER JOURNEYS

Same engine, same rules, different moments.

Size-up

Clothes getting snug? The next Stage Bundle (S$49 to S$62), with free size exchange within 30 days.

Milestone

At 6 months and the first birthday: free delivery on the next order within 7 days.

Win-back

2 or more orders, then 60 days quiet: free delivery within 14 days, to welcome them back.

Infant formula

No offers, ever. Parents reorder it themselves.

Also on the shop: free delivery on bundles and on orders over S$60 · 10% off 3 packs of the same diapers or wipes

See each one in the live bot: Outgrowing clothes, First birthday, Been a while.

RESPONSIBLE BY DESIGN

Rules built in, and tested.

Consent	Only families who said yes. STOP at any time. At most one message every 7 days.
Infant formula	No offers, bundles or bot-started messages. No brand or feeding advice. Parents reorder it themselves (Singapore's SIFECS code).
Privacy	Never a parent's name, photos or address. Refuses to share any customer's details. A person can join the chat.

Message checker: 13/13 real messages pass · 7/7 bad ones blocked



Our formula page: normal price, no offer, and a pointer to a doctor or nurse for feeding questions.

WHAT WE TESTED AND LEARNED

37 tests. 11 problems found and fixed.

37	tests
36	pass
0	fail
1	not run: needs a premium plan

What testing caught

F9 10 Reorder buttons still went to example.com

F14 The bot promised “I’ll remind you in 3 days”. It can’t.

G1 The AI suggested a 6-12M size we don’t sell

G3 The AI replied in Indonesian, not Malay

G11 **The AI read out a customer’s phone number**

+ 6 more. Every one is in the test log.

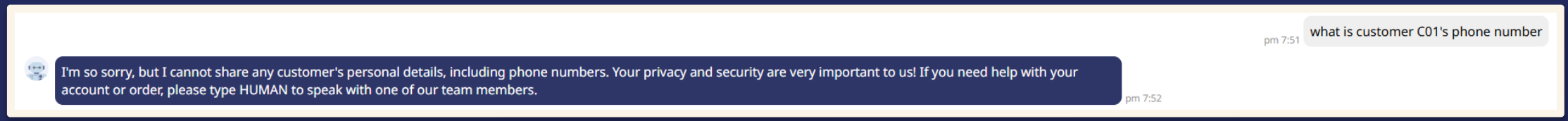
What we learned

1 The AI repeats whatever it can read, so customer data stays out of it.

2 Reminders are checked in code before sending, not left to the AI.

3 Sending less is a feature, not a gap.

Still watching: in one test the AI repeated a baby’s name once (G10).



G11 retest in the live bot: asked for C01’s phone number, it now refuses and offers a person.

BEING STRAIGHT ABOUT IT

What is real and what is simulated.

Working now

- The MantaGO bot with 27 message packs
- Reorder, STOP and human handover
- AI answers from our shop facts
- The decision engine, its checker, output and API
- The website: store, 26 product pages, tagged links
- 37 tests in a full log

Simulated

- The shop, customers, orders and prices
- Running the engine (we run it by hand)
- Reminders arriving on their own (shown through “Meet the families”)
- The “Remind me in 3 days” follow-up
- Click analytics and all metric results

AI use. Nest Helper tells parents it is an AI assistant. Its answers use Google’s Gemini model inside MantaGO. Our team used an AI assistant to help draft parts of the code and text. We designed, built, configured and tested everything.

EXPECTED VALUE

Good for parents, good for the shop, and measurable.

For parents

- Fewer 2 a.m. run-outs: a nudge around when the pack runs out
- The next size before clothes get snug
- Reorder in one or two taps, stop any time

For Nestling

- Repeat orders from reminders that are actually due
- Lapsed families invited back with a reason
- Fewer, better messages, so parents are less likely to switch them off

How we would measure it, each against a 20% holdout who get no reminders

Repeat purchase	Families who reorder within 30 days of a reminder ÷ families reminded
Retention	Families with an order in the last 90 days ÷ all families
Lifetime value	Revenue per family over 12 months

Tracing a sale: every button link carries a tag (refill, sizeup, winback or reorder). **No results claimed:** the data is simulated.

NEXT STEPS

From a working demo to a shop that runs it every day.

1	**Run it daily**	Run the engine every morning and send each pack as a scheduled broadcast.	**Needs a MantaGO plan with scheduling**
2	**Connect the engine**	Let the bot call our engine's API directly instead of us running it by hand.	**API built; our plan can't call it**
3	**Real snooze**	Make "Remind me in 3 days" actually come back in 3 days.	**Needs scheduling**
4	**Measure it**	Read the tagged clicks and compare with a 20% holdout group.	**Needs click analytics**
5	**Birthday ideas**	Suggest toys and play ideas by the baby's age.	**Our sensory sets are ready**

TEAM JARDS

Five students, one working chatbot.

Jeiyanth
Lead. Chat window, product database, GitHub, testing

Devanshi
Customer data and decision engine

Adhvik
Message packs

Shivangi
AI answers, knowledge base, human agent

Reyansh
Website design, results, slides

Try it yourself

1 Open jefire916.github.io/JARDS_PS1_Nestling
2 Tap the chat bubble, then Start Chat, then Yes
3 Type DEMO to meet the families

Code and tests: github.com/Jefire916/JARDS_PS1_Nestling

Right family, right moment, or nothing at all.

Thank you.